TAMIL NADU STATE BOARD - STANDARD 11 CHEMISTRY VOLUME 2

CHAPTER 10: CHEMICAL BONDING

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Chemistry Volume 2
Chapter Name:	Chemical Bonding

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Atomic Orbitals Wave Functions (s & p)

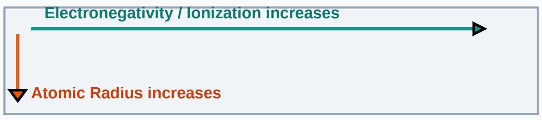
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Octet Rule	Atoms tend to combine so that they each have eight electrons in their valence shell, achieving noble configuration.
Covalent Bond	A chemical bond formed by the sharing of one or more electron pairs between constituent atoms.
Hybridization	The mixing of atomic orbitals of comparable energies to form new degenerate hybrid orbitals.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

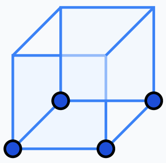
Periodic Properties Trends Direction Map



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Crystal Solid Lattice Unit Cell Model



Simple Cubic Lattice System

Essential Formulas, Laws & Identities:

- Formal Charge: FC = [Valence] - [Lone Pairs] - 0.5 * [Shared]
- Dipole Moment: mu = q * d (SI Unit: Debye)
- Bond Order: BO = 0.5 * (N_b - N_a)

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

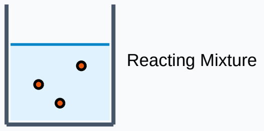
Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Determine hybridization of Carbon in Ethene (C2H4).

Step-by-step Solution:

Carbon forms 3 sigma bonds (two C-H, one C-C) and 0 lone pairs. Steric number = 3. Therefore, hybridization is sp2 (trigonal planar geometry).

Titration Beaker & Solute Precipitation



HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Draw Molecular Orbital energy diagram for Nitrogen molecule (N_2) and calculate its bond order and magnetic nature.

Analytical Solution Strategy:

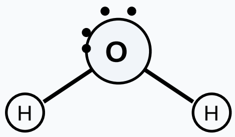
N_2 has 14 electrons: KK sigma(2s)2 sigma*(2s)2 pi(2px)2 pi(2py)2 sigma(2pz)2.

N_b = 8 (excluding KK), N_a = 2.

Bond Order = 0.5 * (8 - 2) = 3 (triple bond).

All electrons are paired, so N_2 is diamagnetic.

Lewis Dot Structure (H_2O Bent Molecular Geometry)



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): State VSEPR theory postulates.

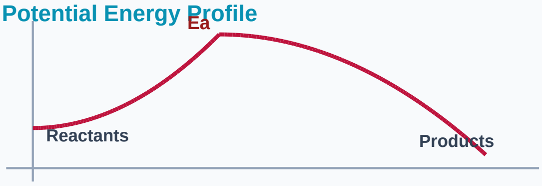
Q2 (2-Mark): Explain the shapes of NH_3 and H_2O molecules using lone pairs.

Q3 (2-Mark): What is a coordinate covalent bond? Give one example.

Q4 (5-Mark): Calculate formal charge of oxygen atoms in Ozone (O_3).

Q5 (5-Mark): Differentiate between sigma and pi bonds.

Chemical Reaction Potential Energy Profile



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Material Design

Diamond vs Graphite properties are mapped using sp3 vs sp2 carbon hybridization configurations.

Application Case: Drug Binding

Pharmaceutical targets are matched to ligands using hydrogen bonding dipole calculations.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Aromatic Benzene Ring (C_6H_6) Resonance

Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.

✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.

✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3, N) for the final numeric answer.

✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

[ ] Read and memorized core definitions on Page 2.

[ ] Reviewed critical formulas and relations on Page 3.

[ ] Solved worked examples on Page 4.

[ ] Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress. Good luck!

Electrochemical Galvanic Cell System